

REVIEW ARTICLE

The usage of Cyanobacteria in wastewater treatment: prospects and limitations

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Significance and Impact of the Study: The treatments of wastewaters by the application of Cyanobacteria, which are photosynthetic prokaryotic microbes have lately developed as a promising choice to conservative biological procedures. Nevertheless, the treatment of various categories of wastes containing organic and inorganic pollutants with Cyanobacteria has attracted considerable research interest recently and gave encouraging results. This methodology presents manifold advantages, for instance decreased operating price since requirements for mechanical aeration is not there due to photosynthesis, CO₂ release is reduced, and the huge budget for chemical fertilizers is avoided since Cyanobacteria metabolize the nutrients present in the wastewaters. Furthermore, the biomass recovered can be used for the production of biofuels. The study will be useful for the scientists working in this area since Cyanobacteria-based water treatment plan is the need of the hour.

Keywords

Cyanobacteria, effluent, nutrients, wastewater.

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Abstract

The applicability of Cyanobacteria executes various roles in the treatment of wastewater, assembling of superfluous food and, thus, produces valued biomass which has various applications. Besides this, they enrich and improve the quality of water as they are photosynthetic autotrophs. Currently, Cyanobacteria gained momentum for remediation of wastewaters because firstly, they enhance the O₂ content of waters through photosynthesis and perform bioremediation of some heavy metals. Secondly, Cyanobacteria play significant roles in distressing the biological oxygen demand, chemical oxygen demand, turbidity, minerals and microbes; thirdly, they can be used either as axenic cultures or as mixed cultures both offering distinct advantages. Lastly, some species are tolerant towards extreme temperatures both low and high, acidic pH, high salt concentrations and heavy metals, which makes them outstanding candidates for the wastewater treatment plants. The suitable immobilization methods must evolve, better understanding of their morphological and biochemical parameters is required for the optimum growth, easy methods of harvesting the biomass after the treatment are required and more trials on large-scale basis are required before they can be launched on full-fledged basis for wastewater treatments.

Introduction

Phytoremediation includes the usage of plants, algae and related microflora for the discharge of contaminants from wastewater (Sood *et al.* 2012; Ali *et al.* 2013; Franchino *et al.* 2013; Richards and Mullins 2013). Oswald and Gotaas (1957a) are the innovators in this field, specifically when it comes to the demonstration of the potency of

algal wastewater treatment. Cyanobacteria-based treatment of wastewater is predominantly effective because it has multifarious outcomes and the capability to treat numerous problems on-site. This review is an effort to recap the information available on the types and diversity of Cyanobacteria utilized in wastewaters and discuss their advantages, future prospects and limitations over other approaches is also discussed.

Cyanobacteria in wastewater treatment

The idea of exploitation of Cyanobacteria in effluent treatment was proposed by (Oswald and Gotaas 1957b) and preliminary trials were performed by Caldwell (1946). Early in 1980, impetus of applying Cyanobacteria in wastewater treatment has grown due to pronounced prospective to take up inorganics. In this concern, Cyanobacteria exhibit an eye-catching substitute, established on the high flexibility of their metabolic machineries, which can be used for bioremediation purposes. Mass cultivation of Cyanobacteria is less costly as well as, more suitable compared with fungi and bacteria. Manipulation of Cyanobacteria in effluent treatment is a current need.

Wastewater treatment systems exploiting photosynthetic microorganisms have lately arose as an encouraging substitute to conservative biological processes (Gonçalves *et al.* 2017). However, the treatment of wastewaters discharged from different locations with Cyanobacteria has collected marked research interest over the last 10 years and gave encouraging output in both organics and inorganics discharge (Subramaniyam *et al.* 2016; Ferreira

et al. 2017; Papadopoulos *et al.* 2020a, 2020b). This process offers many benefits, like decreased operational cost as photosynthesis substitute mechanical requirements for aeration, decreased CO₂ releases and the high cost of using chemical fertilizers for microalgal growth is prevented as microbes use the nutrients present in the wastewaters (Ghimire *et al.* 2017). Moreover, the biomass that is accumulated can be used as a resource for the production of biofuels (Ganesh Saratale *et al.* 2018; Ghimire *et al.* 2017; Koutra *et al.* 2018) (Fig. 1). Cyanobacterial biofilms also display great prospective in wastewater treatment uses. The biofilm matrix defends microbial inhabitants from external environmental stresses (Liu *et al.* 2017) (Table 1).

Advantages over other microorganisms

The photoautotrophic character and capacity of some species to fix atmospheric nitrogen enable Cyanobacteria to be producers, as opposed to consumers, and make their cultivation cheap (Hensyl 1989). Cyanobacteria are also known to grow densely in polluted locations where,

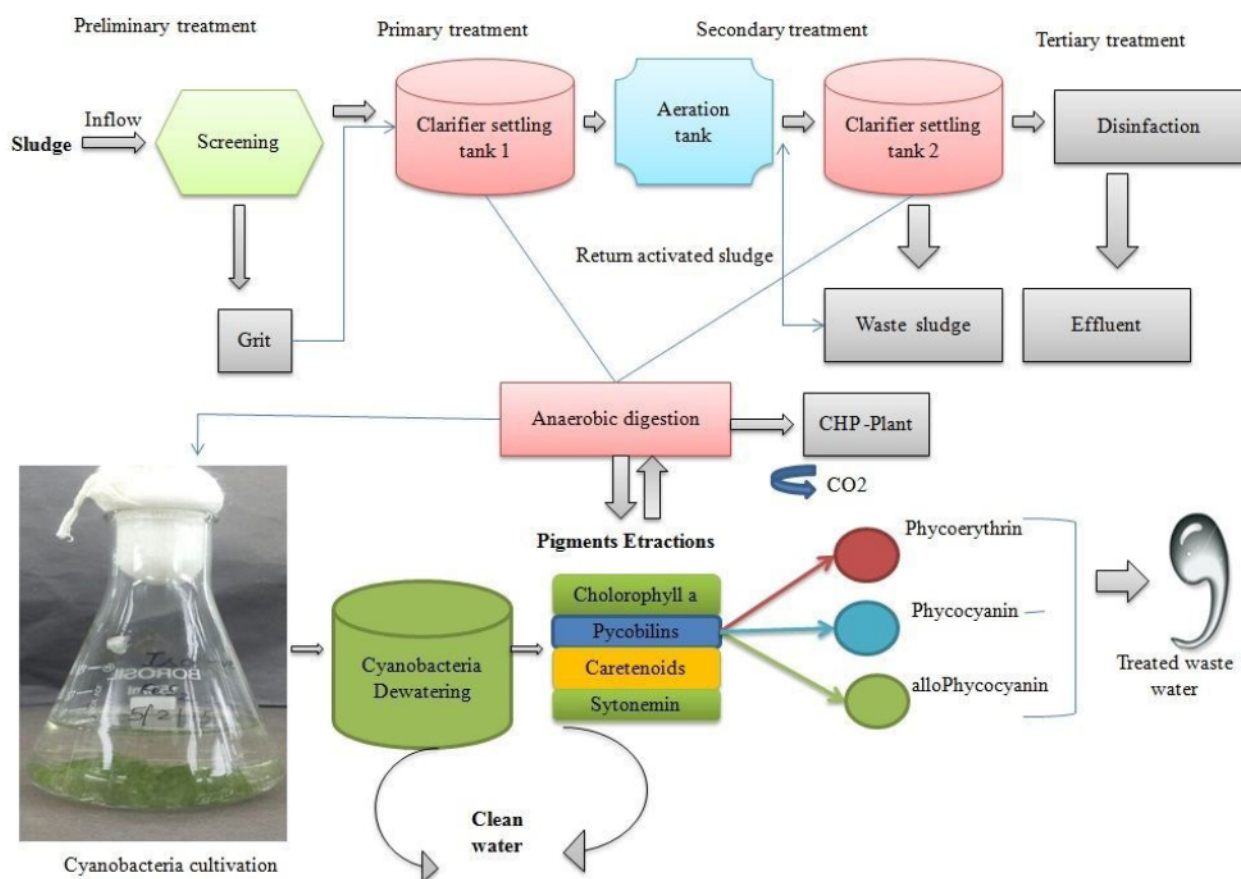


Figure 1 The wastewater treatment process using Cyanobacterial system.

Table 1 Cyanobacterial strains in wastewater treatment

Species	Reactor	Effluent	Reduced nutrients	COD	BOD	References
<i>Nostoc muscorum</i> <i>Anabaena subcylindrica</i>	Conical flasks	Sewage and industrial wastewater effluents	PO ₄ ³⁻ —(20.8–95%) NO ₃ ⁻ —(19.6–80%) NH ₃ —(20.9–96%)	20–57.1%	N/A	El-Sheekh <i>et al.</i> (2011)
<i>N. muscorum</i> , <i>Anabaena variabilis</i>	Batch system	Mixed domestic-industrial influents	N/A	32.14% and 35.71%	54.55 and 56.36%	Talukder <i>et al.</i> (2015)
<i>Anabaena oryzae</i>	Glass Vessels	Domestic-industrial Wastewater	N/A	73.68%	589.29%	El-Bestawy (2008)
<i>A. oryzae</i> , <i>A. variabilis</i>	Wolf [®] bottles	Wastewater effluent of sewage station	Zn—86.12% Cu —94.63%	73.6%	89.29%	El-Bestawy (2008)
<i>Spirulina platensis</i>	Bioreactors	Municipal drain	N/A	N/A	N/A	Converti <i>et al.</i> (2006)
<i>S. platensis</i> , <i>Chlorella vulgaris</i>	Mini reactors	Sludge water	N/A	N/A	N/A	Jalal <i>et al.</i> (2011)
<i>Geitlerinema sp.</i> , <i>Gloeotrichia sp.</i> and <i>Lyngbya sp.</i>	Flask	Shrimp farm wastewater	Total ammonia nitrogen —98 ± 0.05, 93 ± 0.02, 94 ± 0.01 Total nitrite nitrogen —93 ± 0.04, 83 ± 0.05, 80 ± 0.04 Total soluble reactive phosphorous—74 ± 0.03, 56 ± 0.02, 59 ± 0.01	N/A	6.59 ± 0.06	Banerjee <i>et al.</i> (2015)
<i>Microcystis aeruginosa</i>	Laboratory incubator	Raw water and bloom water	Al—705 ± 26.4 Fe—32.6 ± 4.2	9–12 mg l ⁻¹	N/A	Xu <i>et al.</i> (2016)
<i>N. muscorum</i> , <i>A. variabilis</i> , <i>Lyngbya majuscula</i> and <i>Oscillatoria salina</i>	Erlenmeyer flasks	Sewage discharges	Ni—8.4 to 2 (mg l ⁻¹) Zinc —12.1 to 3.3 mg l ⁻¹ Ca—108 to 78.1 mg l ⁻¹ Mg—78 to 50 mg l ⁻¹	459.3–172.6 mg l ⁻¹	104–21.5 mg l ⁻¹	Noel and Rajan (2017)
<i>Tolypothrix ceytonica</i>	Glass Vessels	Domestic-industrial wastewater	Zn—86.12% Cu—94.6%	51%	280%	El-Bestawy (2008)
<i>Cyanophyceae</i>	Reactor	Dairy sewage water	Nitrate—0.639 mg l ⁻¹ NO ₃ ⁻ —675 mg l ⁻¹	88 mg l ⁻¹	97.3 mg l ⁻¹	Bernal <i>et al.</i> (2008)
<i>Aulosira fertilissima</i>	Batch system	Fish pond discharge	NO ₂ ⁻ —3.2 ± 0.17 to 0.6 ± 0.02 mg l ⁻¹ NO ₃ ⁻ —12.2 ± 0.49 to 13.1 ± 0.45 mg l ⁻¹ NH ₄ ⁺ —2.1 ± 0.11 to 0.8 ± 0.01 mg l ⁻¹ PO ₄ ³⁻ —2.8 ± 0.21 to 1.1 ± 0.05 mg l ⁻¹	238.6 ± 3.16 to 99.3 ± 1.39 mg l ⁻¹	113.9 ± 1.52 to 59.3 ± 0.99 mg l ⁻¹	Samantaray <i>et al.</i> (2011)
<i>Oscillatoria annae</i>	Erlenmeyer flasks	Tannery industry effluent	Nitrate—86.48% Nitrite—23.5% Calcium—63%	N/A	114–43.67 mg l ⁻¹	Shankar <i>et al.</i> (2013)

The usage of different Cyanobacterial species in the treatment of wastewater.

they are widely distributed and dominate the micro floral populations (Fay and Van Baalen 1987). They acquire natural resistance and selectivity against environmental pollutants due to their presence in the polluted systems. However, their sustainability and metabolic activities are not embellished by reducing levels of biodegradable pollutants. They have been efficiently used as a low-cost method for remediating dairy wastewater (Lincoln *et al.* 1996), dissolved inorganic nutrients from fish farms and nutrients (N and P) from sewage effluents (Hu *et al.* 2000; Phang *et al.* 2000; de-Bashan *et al.* 2004). Many Cyanobacterial strains have been completely sequenced and studied with an intention of standardizing their constructive use for reclamation of wastewaters.

Limitations of Cyanobacteria-based wastewater treatment

The most prominent difficulty in the application of Cyanobacteria for wastewater treatment is their harvest and revival from the treated effluent (Chevalier and de la Noüe 1985). Immobilization seems to provide many advantages in compared with batch or continuous fermentation systems in which where free microorganisms are used (Hall and Rao 1989). All the Cyanobacterial strains are not similarly effective for wastewater treatment. Their efficacy in wastewater treatment relies on mainly on two chief factors: (i) the ability of the strains to propagate under the prevailing ecological conditions and (ii) its efficiency to discharge the contaminants. The operation and efficiency of the wastewater treatment units situated in the areas with harsh ecological conditions relies upon the choice of the microbes that can grow under the specific harsh environmental conditions (Talbot *et al.* 1991; de-Bashan *et al.* 2004). Therefore, Cyanobacterial strains intended for degradation of pollutants must pass the adaptability test to the local wastewater scheme (de-Bashan *et al.* 2008).

Nostoc muscorum and *Anabaena* spp.

The two Cyanobacterial strains, namely, *Nostoc muscorum* and *Anabaena subcylindrica* have been extensively applied for assessing the extent of pollution caused by industrial effluents (El-Sheekh *et al.* 2011). In another study the amounts of magnesium, calcium, sulphate, nickel and zinc showed reduction after treatment with Cyanobacterial consortium. The comparison between the two species showed higher efficacy of *Anabaena variabilis* in reducing the pollution load than *N. muscorum* from textile industry effluent (Talukder *et al.* 2015).

Nostoc muscorum, *Anabaena variabilis*, *Lyngbya majuscula* and *Oscillatoria salina*

The Cyanobacterial consortium is reported in the bioremediation of effluent from textile industry. The given

study reports the uses of four species of Cyanobacteria in the removal of various contaminants and for the treatment of wastewater. In the present study, effluent samples were obtained from textile industry located in Chinnalappatty, Dindigul district, Tamil Nadu, India, and incubated with mixed Cyanobacterial culture including *N. muscorum*, *A. variabilis*, *Lyngbya majuscula* and *Oscillatoria salina* in BG11 media and contaminated effluent was added for bio treatment. A marked increase in pH value and declination in COD, BOD, Ca, Mg, sulphate, zinc, nickel and colour were seen after 25 days of treatment (Noel and Rajan 2017).

Anabaena oryzae, *Anabaena variabilis* and *Tolypothrix ceptonica*

The objective of the study was to investigate the potential of the main Cyanobacterial species controlling the lake ecosystem towards the biodegradation and biosorption of organic and inorganic pollutants after the primary treatment of effluents. The effluents were subjected to three axenic Cyanobacterial cultures for biological treatment applied as batch system. Discharge efficiencies of the various pollutants were calculated and equated. The elimination of the pollutants by Cyanobacteria showed the species and contaminant-dependent removal. The highest efficacies were obtained by *A. variabilis* and *Anabaena oryzae*. The maximum discharge of the total solids reported was shown by *Tolypothrix ceptonica*, followed by *A. variabilis*. In case of pollutant metals, *T. ceptonica* exhibited the highest biosorption capacity (El-Bestawy 2008).

Spirulina platensis

The efficacy of a bench-scale tubular photobioreactor for batch culture of *Spirulina platensis* was evaluated under limited light conditions. The usage of airlift equipment for the re-cycling of biomass instead of conventional pumps is the most striking feature of this reactor to avoid trichome damage owing to mechanical stress. In this plant, a highest cell concentration was achieved after 15 days of cultivation. The experimental biomass concentrations attained under different conditions were higher than those reported earlier (Converti *et al.* 2006).

In another study, the combination of Cyanobacteria and green alga also showed good response towards the treatment of wastewater, that is, *S. platensis* and *Chlorella* sp. in different conditions. High growth, nitrogen and phosphorous uptake rates were recorded with *Chlorella* on centrate (Jalal *et al.* 2011). Co-digestion with *S. platensis* increased dewaterability of biosolids; however, *Chlorella* had a little negative impact on dewaterability compared with WAS alone. Both high ammonia concentrations and mutual self-shading partially inhibited the

growth of *S. platensis* due to the high biomass density in the airlift reactor (Yuan *et al.* 2012).

Oscillatoria spp.

The application of *Oscillatoria*, *Nostoc*, *Synechococcus*, *Nodularia* and *Cyanothece* was studied by Dubey *et al.* (2011) for bioremediation of tannery effluents. The Cyanobacterium was effective in decolorizing the effluent and the maximum growth was recorded at the optimum temperature $29 \pm 2^\circ\text{C}$ and pH near 7 (Shankar *et al.* 2013). Scientist recorded the bioremediation of polluted water in removal of components like PO_4^{-2} , NO_3^{-2} , SO_4^{-2} and Cl^- by different species of *Oscillatoria* and *Nostoc* (Azarpira *et al.* 2014). They were highly efficient in removing the NO_3^{-2} , PO_4^{-2} , SO_4^{-2} , Cl^- . The prospective use of *Oscillatoria okeni* and *Chlorella vulgaris*, to remove nitrate-nitrogen from tilapia-pond effluent was also reported (Attasat *et al.* 2013).

Phormidium spp.

In a study, chitosan: *Phormidium* combinations were used as a biological treatment for the removal of nitrogenous and phosphorus compounds from a secondary effluent in a batch process. 71 and 92% of phosphates were removed after 6 and 24 h, respectively. The orthophosphate removal rate was identical for all three concentrations of algae-chitosan tested and showed 90% removal (de la Noüe and Proulx 1988).

The biological treatment of domestic wastewaters was studied to find out the prospects and limitations of the epilithic filamentous *Phormidium bohneri* (Talbot and de la Noue 1993). A complete removal of NH_3 and PO_3^- was obtained after 3 and 5 days, respectively at 30°C by the cultures of *P. bohneri* obtained after anaerobic digestion of swine manure (de la Noüe and Basseres 1989). The removal of nitrogen from polluted waters using bench-top bioreactors incorporating the thermophilic Cyanobacterium, *Phormidium laminosum* has also been stated (Garbisu *et al.* 1992).

In another study, four strains of filamentous, mat-forming Cyanobacteria isolated from Arctic and Antarctic environments were assessed for the development of tertiary biological wastewater treatment systems and a tropical strain, *P. bohneri*. The culture medium mimicked real effluent conditions having high amount of nitrate and phosphate (Chevalier *et al.* 2000).

Another study was undertaken in which 49 cultures of filamentous, mat-forming Cyanobacteria were purified from the Arctic, subarctic and Antarctic ecologies. They were screened for their probable application in open air wastewater treatment plans made for cold north-temperate climates. The best isolate, strain E18 (*Phormidium* sp.) from a high Arctic lake showed good growth at

low temperatures and formed clumps and could be easily recovered by sedimentation. Polar Cyanobacterial strain like E18 are suitable for treatment of wastewater in cold weathers (Tang *et al.* 1997). In the current research, biological treatment of wastewater by five isolates of Cyanobacteria was explored separately under hot climatic conditions. The marine isolate of *Phormidium* was the most effective strain amongst all the strains tested. Biomass production in wastewater was comparable with that of cultural media (Das and Santra 2010).

Geitlerinema sp.

The removal efficiency of nitrogenous compounds and phosphorus by three periphytic Cyanobacterial mats (*Geitlerinema* sp., *Gloeotrichia* sp. and *Lyngbya* sp.) immobilized to polyvinyl chloride sheets by self-adhesion were screened. It was found that *Geitlerinema* sp. showed the maximum reduction rate (98%), and wastewater concentrations and total ammonia nitrogen, nitrite nitrogen and soluble reactive phosphorus were reduced (Banerjee *et al.* 2015).

Microcystis aeruginosa

Microcystis aeruginosa was studied by Xu *et al.* (2016). The result indicated that viability of *M. aeruginosa* in PAFC sludge was better than that of cells in AlCl_3 or FeCl_3 sludge alone after the same storage time. In AlCl_3 and FeCl_3 sludge high amounts of Al and Fe were lethal to *M. aeruginosa*, whereas in PAFC sludge low levels of Al showed slight toxic effect on *M. aeruginosa* growth and moderate amounts of Fe were beneficial to growth. Although the cell viability in FeCl_3 sludge was less as in case of AlCl_3 sludge, the *M. aeruginosa* cells stayed unbroken after 10 days of storage time (Xu *et al.* 2016).

Lyngbya

Marine strain of Cyanobacterium *Lyngbya* sp. was studied for its biodegradation potential of textile dye discharge. Groundnut shell was used as an adsorbent in this study. 77.55% discoloration was obtained in the textile discharge after treatment with *Lyngbya* sp. The heavy metals like zinc, mercury, nickel, cadmium, chromium and iron existent in textile waste were reduced and adsorbed on the surface *Lyngbya* sp. filament (Nandhini *et al.* 2014).

Aulosira fertilissima

The study was done with the objective to develop a natural recirculatory wastewater treatment process with a diazotrophic Cyanobacterium, *Aulosira fertilissima*, and simultaneous creation of valued product in the form of poly- β -hydroxybutyrate (PHB). Maximum accumulation of PHB was seen in *A. fertilissima* in sedimented fish

pond discharge with stirring. Under standard conditions, the PHB yield was increased for all the seasons. Analysis of the data indicated that a hectare of *A. fertilissima* cultivation in fish pond discharge would give an annual harvest of ~17 t dry biomass, comprising of 14 t of PHB (Samantaray *et al.* 2011).

Leptolyngbya sp.

The recent study presented a new Cyanobacteria-based biofilm photobioreactor and evaluates its results after the brewery wastewater treatment on production of biomass. Two supporting materials, that is, glass and polyurethane were studied to explore the consequence of superficial hydrophobicity on biomass attachment and total performance of the bioreactor. The reactor showed high discharge efficacy of the wastewater's contaminants, while biomass per reactor surface was also good (Papadopoulos *et al.* 2021). In another study, brewery wastewater was treated by applying a consortium of Cyanobacteria and bacteria predominantly containing filamentous Cyanobacterium *Leptolyngbya* sp. (Papadopoulos *et al.* 2020).

Consortium of different species of Cyanophyceae and Chlorophyceae

In the current study, dynamics related to water quality alterations by a consortium of Cyanobacteria and green microalgae was undertaken in municipal wastewater that receive effluent from the dairy industry in central Veracruz, Mexico, by an experimental stabilization-reactor system. Thirty-one strains of Cyanobacteria were used and 15 physicochemical parameters were assessed daily for 25 days in pretreated water. The biomass of Cyanobacterial groups showed marked negative correlations with BOD, COD and total suspended solids. In contrast, a marked positive correlation was observed with DO, pH and nitrates. Using the Canonical Correspondence Analysis an initial anaerobic phase (Phase I) was recognized, studied by high BOD, TSS and COD, ammonium and faecal coliforms. This was followed by a transition phase (Phase II), an initial aerobic phase (Phase III), when water quality enriched considerably, and finally an aerobic phase (IV) with a high percent reduction in these parameters and almost full removal of organic load. The most plentiful species linked with these phases were *Arthrospira jenneri* (Cyanophyceae) and *Coccomonas* sp. (Chlorophyceae) in Phase I; *Polytomella* sp. (Chlorophyceae) in Phase II; *Polytomatetraolare*, *Chlamydomonas caeca* (Chlorophyceae) and some Cyanophyceae including *Geitlerinema*, *Synechocystis*, *Cyanobium* and *Glaucospira* in Phase III; *Carteria* sp. (Chlorophyceae), *Lepocynclis ovum* and *Euglena clavata* (Euglenophyceae) in Phase IV. Maximum richness, diversity, and biomass levels occurred during the aerobic phase (Bernal *et al.* 2008).

Cyanobacterial diversity

The existence of nutrients particularly nitrogen and phosphorus, in the form of nitrate, nitrite, ammonia or phosphorus in wastewater results into eutrophication (Liu *et al.* 2010). Cyanobacteria create an expansive group of organisms including which have the characteristic of photoautotrophic metabolism and are dispersed both in fresh and marine waters, with a broad range of variety in their organization and habitat (Lee 2008). Cyanobacteria constitute an important part of the microbial diversity of wastewaters, which can also show a part in the self-purification of these wastewaters (Sen *et al.* 2013). The biodiversity of Cyanobacteria is tremendous and the best model (Gompertz) estimated that this group contains 6280 different species. This huge diversity and tendency to adapt to harsh and difficult environments has paved the way for the scientific community to screen, identify potential strains and to develop promising Cyanobacteria-based methodologies for wastewater treatment (Fouilland 2012).

Production of cyanotoxins and its implications in wastewater treatment

Cyanobacterial cells proliferate in municipal wastewater treatment facilities (WWTFs) which flourish in steady, nutrient-rich environments and favoured by the extended water residence times (Amengual-Morro *et al.* 2012; Chu *et al.* 2018). Dense Cyanobacterial blooms may disturb the supply of recycled water because of the synthesis of extracellular polymeric compounds and also affect micro-filtration process by adhering to the surface of the filtration units (Kwon *et al.* 2005). Some strains of Cyanobacteria produce toxins, which might pose a health risk to the public (Daly *et al.* 2007). In such a case, it is very important to use different techniques to identify and monitor Cyanobacterial growth and their toxins in WWTFs. These techniques include microscopy, ELISA, chemoanalytical methods, flow cytometry and molecular methods. The focus should be on techniques for the management of site-specific Cyanobacterial blooms. Application of omics for long term might help in the identification of indicator species and the formulation of site-specific bloom dynamic models for the alleviation and management of Cyanobacterial blooms in WWTFs. The newly developed metagenomic tools could potentially provide deep understanding of the biodiversity and flux of toxin-producing Cyanobacterial species in these systems; they might be considered a complement to, quantitative chemoanalytical methodologies.

In the particular strategies for wastewater management, the growth of non-toxic Cyanobacterial species is

intentionally encouraged to expedite bioremediation and to produce biomass for biofuel production (Korosh *et al.* 2018).

Traditionally *Phormidium*, a non-toxic species has been applied for the removal of nutrient during tertiary treatment (Chevalier *et al.* 2000). Other Cyanobacterial genera have also been found to proliferate during different stages of treatment in WWTFs including *Microcystis* (Shanthala *et al.* 2009), *Pseudoanabaena* (Soh Kengne *et al.* 2014), *Lyngbya* (Kumar and Goyal 2020), *Oscillatoria* (Badr *et al.* 2010), *Limnothrix*, *Nostoc* (Furtado *et al.* 2009), *Synechocystis* (Oudra *et al.* 2000), *Aphanocapsa* (Florentino *et al.* 2019), *Merismopedia* (Sutherland *et al.* 2017) and *Synechococcus* (D'Alessandro *et al.* 2020).

However, the occurrence of Cyanobacteria in wastewater treatment plans is not essentially suggestive of the presence of Cyanobacterial toxins. The possibility for toxicity is dependent on the strain (Christiansen *et al.* 2008; Srivastava *et al.* 2016). Even if the required genes of toxin biosynthesis are present, toxin synthesis can be provisionally stopped due to unfavourable environmental conditions (Wood *et al.* 2011). The common factors which regulate the production of cyanotoxin include intensity of light and exposure time, concentrations of nutrient, temperature of water and stratification, and also the production of allelochemicals by competing organisms (Neilan *et al.* 2013).

The presence of cyanotoxins in wastewater can severely affect public health as one can get exposure through several routes, which includes agricultural and industrial applications as a result of which consumption of toxic recycled water can occur, or the discharge of recycled water into wetlands and recreational waterways.

Several strategies can be adopted for monitoring toxin-producing Cyanobacteria in WWTFs, which involves the combinatorial assessment of several biological, biochemical and physicochemical parameters, enumeration and biovolume calculations for the assessment of toxicity and microscopic Cyanobacterial cell counting and biovolume calculation (Lawton *et al.* 1999; Joung *et al.* 2006). Amplicon-based next-generation sequencing (NGS) method for monitoring Cyanobacteria is an advanced technology used for this purpose. The identification of Cyanobacteria is done using Sanger sequencing, which depends on the isolation axenic culture, followed by the PCR amplification of a suitable taxonomic marker region and finally sequencing through chain termination PCR (Rego *et al.* 2019). Shotgun metagenomic approaches for NGS monitoring Cyanobacteria in WWTFs involving the sequencing of all genes in all organisms present in a given environmental sample is a newer technique for the identification species and strains, and detecting of biodiversity (Carr and Borenstein 2014; Campanaro *et al.* 2016; Ranjan *et al.* 2016). The microbial gene expression might also

be monitored by applying metatranscriptomics and metaproteomics in WWTF. Metatranscriptomics discusses the study of total RNA expressed within a system at a given time point. It may be used to observe the diversity and activity of genes under given conditions. In the same manner, metaproteomics is the study of total proteins expressed within a system at a given time point, including structural proteins and enzymes of primary metabolism.

Degradation of pollutants by Cyanobacteria in Erlenmeyer flasks using the concept of a bioreactor

The treatment and removal of benzene, toluene, ethylbenzene, xylenes, styrene and isopropylbenzene (BTEXSI) coming from six wastewater treatment plants of textile industries has been studied. The data showed that BTEXSI were abundant in the textile dyeing wastewater, whereas isopropylbenzene, toluene and xylenes were mainly present in raw wastewaters (Ning *et al.* 2015). BTEX (benzene, toluene, ethylbenzene and isomers of xylene) compounds are the major pollutants present in the wastewater that is discharged into water bodies during petroleum extraction. In a study the capacity of peat and angico hardwood sawdust was investigated to remove these compounds from the wastewater during petroleum extraction (Costa *et al.* 2012; Viotti *et al.* 2015).

Another concern is the extensive usage of pesticides in the recent times which poses serious threat to health as they contaminate water (Zaror *et al.* 2010). Numerous types of pesticides have been noticed in surface water and roused pronounced public concern as the pesticides can result into adverse health hazards even at less concentration (Plakas and Karabelas 2012).

Keeping in mind, the above-mentioned pollutants of wastewater coming from different industries, the present experiments were designed. The Cyanobacterium used in the study was *N. muscorum*, which is filamentous and heterocystous species. The axenic cultures of Cyanobacteria were maintained on the medium as control. In the test cultures, the different pollutants were added in the medium at the time of inoculation. The GC analysis of the pollutants was carried out when the cultures were grown to log phase. The amounts of pollutants were estimated in the medium to find out the remaining quantity of these pollutants in order to study the potential of *N. muscorum* to degrade these compounds.

MATERIALS AND METHODS

Maintenance of Cyanobacterial cultures

The axenic cultures of *N. muscorum* was allowed to grow in the nitrogen free BG-11 medium buffered to pH 7.5

with 10 mmol l⁻¹ NaOH. After every 28 days sub-culturing was done by transferring algae to fresh medium under the same conditions to maintain them in the exponential phase. Stock cultures were maintained on solid agar slants. The cultures were raised in bulk, and the 11-day-old cells were harvested and centrifuged; the harvested cells were washed with nitrogen-free medium and resuspended in the same fresh medium. This was used as inoculum for all the experiments. The test cultures were grown in 100 ml of BG-11 media amended with 100 mg l⁻¹ of individual benzene, toluene and xylene (BTX) in an ml Erlenmeyer flask and placed on a shaking incubator (120 rpm) at temperature of 25°C treated with different pollutants which included BTX (isobenzene, isotoluene and isoxylene), pesticides (cypermethrin and dimethoate) and herbicide (butachlor). Pesticide and herbicides used for this research work were purchased from the market keeping in mind all the safety parameters.

Growth behaviour of Cyanobacterial cultures under various pollutants

Evaluation of growth behaviour of *N. muscorum* cultures grown in media supplemented with various pollutants was done by taking whole cell absorbance ($A_{660\text{nm}}$) after every alternate days and by plotting a graph of absorbance versus days.

Gas chromatography of the pollutants

The analysis of the pollutants was done by gas chromatography. An aliquot of 1 ml of the sample was injected in Gas Chromatograph (Clarus 500, Perkin Elmer, Waltham, MA) with (solid-phase microextraction (SPME) sampling system. The reagents used in this work were of analytical grade. SPME sampling involves the exposure of a coated fused silica fibre to the sampling source. Fibre used was 100 micrometre PDMS fibre. 100 μ l of 50 μ g ml⁻¹ STD spiked in 900 μ l Milli Q water (1 ml) and the head space SPME extraction takes place for 30 min at 60°C. Same procedure was followed for all the samples. 100 μ m of PDMS fibre was used in the process. The silica fibre is exposed to the sample in the SPME sampling system. The analytes were extracted by the fibre from the sample. The quantity of the analyte extracted by the fibre is proportional to its concentration in the sample. The SPME fibre was then transferred for separation. Here, desorption of the analyte took place which was analysed by GC. Each component produced a specific spectral peak that was recorded electronically. The size of the peaks is proportional to the quantity of the corresponding substances in the specimen analysed.

Results and discussion

Growth behaviour of Cyanobacterial cultures in the presence of various pollutants

The proper growth of the Cyanobacterial strain used in the study is the essential necessity for the biodegradation of the pollutants. The growth of *N. muscorum* on BG-11 media supplemented with individual BTEX, herbicides and pesticides in Erlenmeyer flask at 25°C has been shown in Fig. 2. *Nostoc muscorum* exhibited almost the same growth pattern in the presence of individual BTX. In the initial phase they were in their lag phase for 2 days as there was no gradual increase in the optical density. The growth started increasing, and the cultures attained log phase after 2 days of incubation till seventh day and gradually entered the stationary phase from the eighth day, and the steady state was achieved further. The overall growth was hampered on BTEX, suggesting the high toxic concentration of BTX. *Nostoc muscorum* showed high sensitivity towards herbicides and pesticides as the cultures growth was less compared with control. Cypermethrin showed least toxicity and there was a delayed log and stationary phases in the presence of butachlor. The growth was highly suppressed in case of dimethoate till 14th day, but there was a rapid increase in growth after that which may be due to better adaptation of cells.

Degradation of pollutants analysed by gas chromatography

The pollutant tolerant and adapted strains were analysed to determine their rates of BTX, herbicides and pesticides degradation. Samples (40 ml) were removed after 21 days, and the degradation of pollutants was analysed by gas chromatography as previously described (Borah and Yadav 2017) also (Madhavi 2014). The degradation of different pollutants showed typical biodegradation kinetics for *N. muscorum* cells grown on individual as media supplement (Table 2). As observed in Fig. 2, the culture used BTX for cell growth as it became a secondary carbon source and BTEX degradation occurred as cells grew in the flask. *Nostoc muscorum* was found to be degrading 99.8, 99.7, 99.1% of benzene, toluene and xylene, respectively after 21 days. This may be due to the utilization of these compounds as secondary carbon source as has been already reported in earlier study. The degradation percentage of butachlor, cypermethrin and dimethoate were 99.8, 99.9 and 78.6%, respectively.

GC analysis is a clear indicator that Cyanobacteria have the potential to be used as potent organisms in the wastewater treatment plans. The herbicides, pesticides and BTX compounds have been degraded to a good extent by

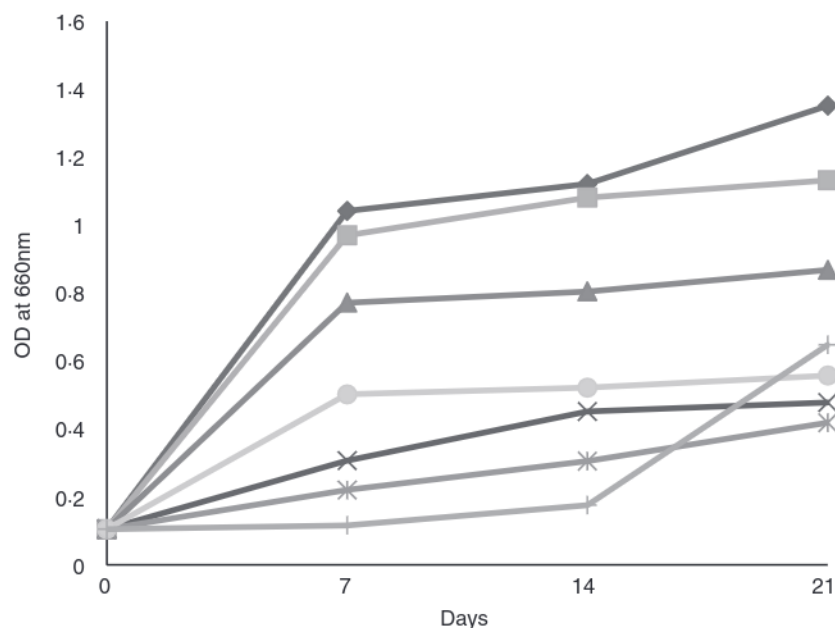


Figure 2 Growth behaviour of *Nostoc muscorum* cultures grown in the presence of different pollutants (♦) control, (■) benzene, (▲) toluene, (x) xylene, (✱) butachlor, (●) cypermethrin, (+) dimethoate.

Table 2 Amount of pollutants detected in samples

S. no.	Pollutant	Amount of pollutant added to the medium ($\mu\text{g ml}^{-1}$)	Amount of stress detected after 25 days of inoculation ($\mu\text{g ml}^{-1}$)
1.	Benzene	100	0.123
2.	Toluene	100	0.089
3.	Xylene	100	0.232
4.	Butachlor	100	0.130
5.	Cypermethrin	100	0.05
6.	Dimethoate	100	21.40

N. muscorum. It was highly effective in the degradation of cypermethrin followed by butachlor and dimethoate. In case of BTX, *N. muscorum* exhibited high potential to degrade benzene followed by toluene and xylene.

Conclusion

The Cyanobacteria find potent applications in wastewater treatment as is evident by several reports, and their role has been well acknowledged. The photosynthetic and autotrophic mode of nutrition in Cyanobacteria make them appropriate candidates for treating wastewater giving extra advantage as they can also enrich the waters along with the treatment. However, some more detailed

and large-scale field trials are required before they can be used full-fledged in water treatment plants. The separation of Cyanobacterial biomass from treated water discharge is another factor which needs to be addressed otherwise it may cause eutrophication. Cyanobacterial biomass production along with biological wastewater treatment can have a better commercial prospect. Some strains of Cyanobacteria are extremophiles and can tolerate harsh conditions like extremely high and low temperatures, highly acidic pH, high concentrations of salts. But, because they are autotrophic, they require good amount of light radiations and proper aeration for the optimum growth and performance. The application of thermophilic Cyanobacteria in wastewater purification has advantages because pollution can be evaded as they are tolerant to high temperature (Sawayama *et al.* 1998).

It is also seen that Cyanobacterial consortium using mixed cultures have given better results in some studies compared with single culture. But, in some cases, it was seen that the axenic culture was better than the mixed cultures, and this may be due to the competition between mixed cultures for nutrients. More research is desired answer these questions and to prepare reusable immobilized particles containing Cyanobacteria for making a cost-effective system. For this, appropriate immobilization methods needs to be explored including co-immobilization of various species are required to be done for symbiotic interaction among them which will result in synergetic enhancement of removal capabilities.

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Conflict of Interest

No conflict of interest declared.

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